Nuclear Energy

Chapter 4

There are many types of energy that people can use to meet their energy needs. One of them is **nuclear energy**. Like other sources of energy, nuclear energy has costs and benefits.

Big Question

What is nuclear power?

About twenty percent of the electricity generated in the United States is provided by nuclear energy. That electricity can provide power to many homes, businesses, schools, and other places.

Vocabulary

nuclear energy, n. energy produced by the splitting of atoms, tiny particles of matter

Materials for Releasing Nuclear Energy

Nuclear energy is generated when atoms of certain materials are split. This breaks the bonds holding the nucleus of the atom together, generating different atoms and releasing energy. The most common material to use for this process is uranium.

Less than one percent of the uranium found on Earth is the kind that can be used for nuclear energy. Uranium is found in rock usually mixed with other elements and even other kinds of uranium that cannot be used in nuclear reactions.

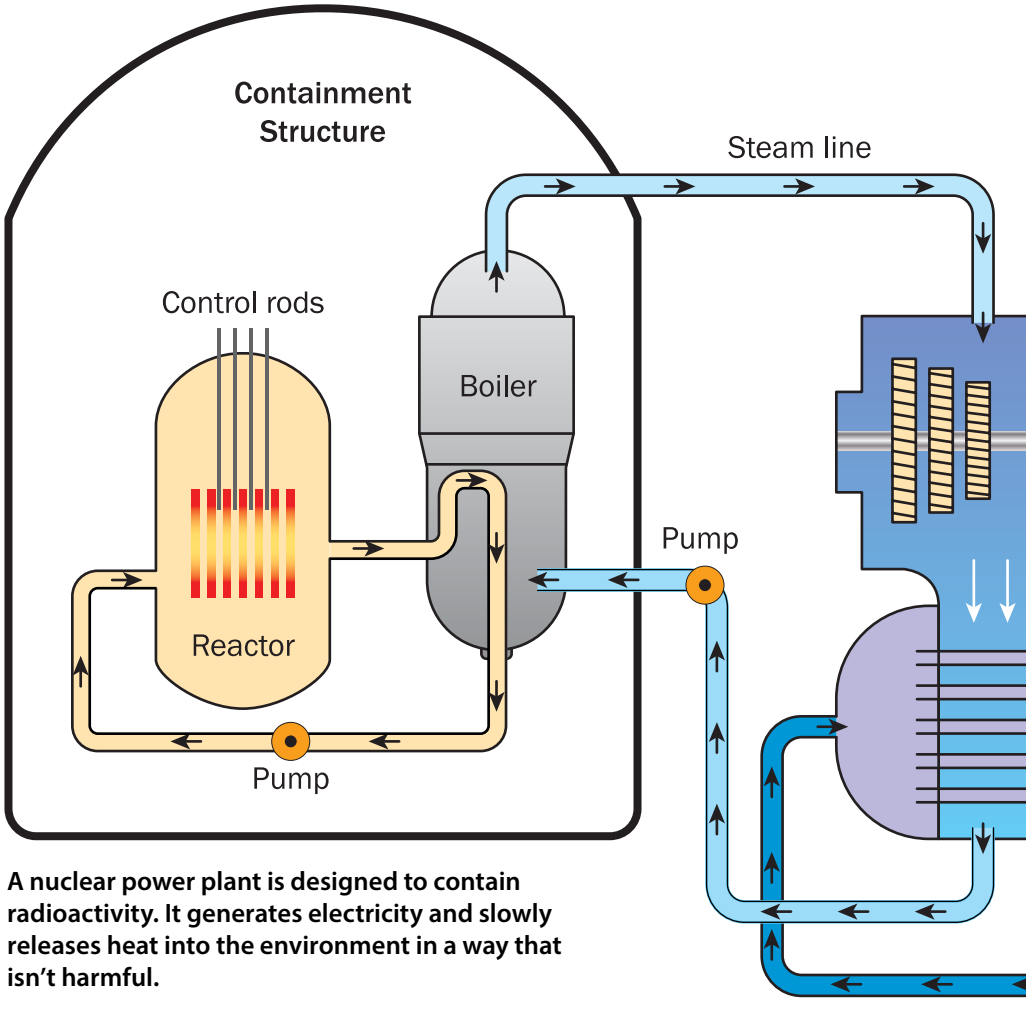


Uranium has to be mined and then enriched.

Nuclear Power Plant

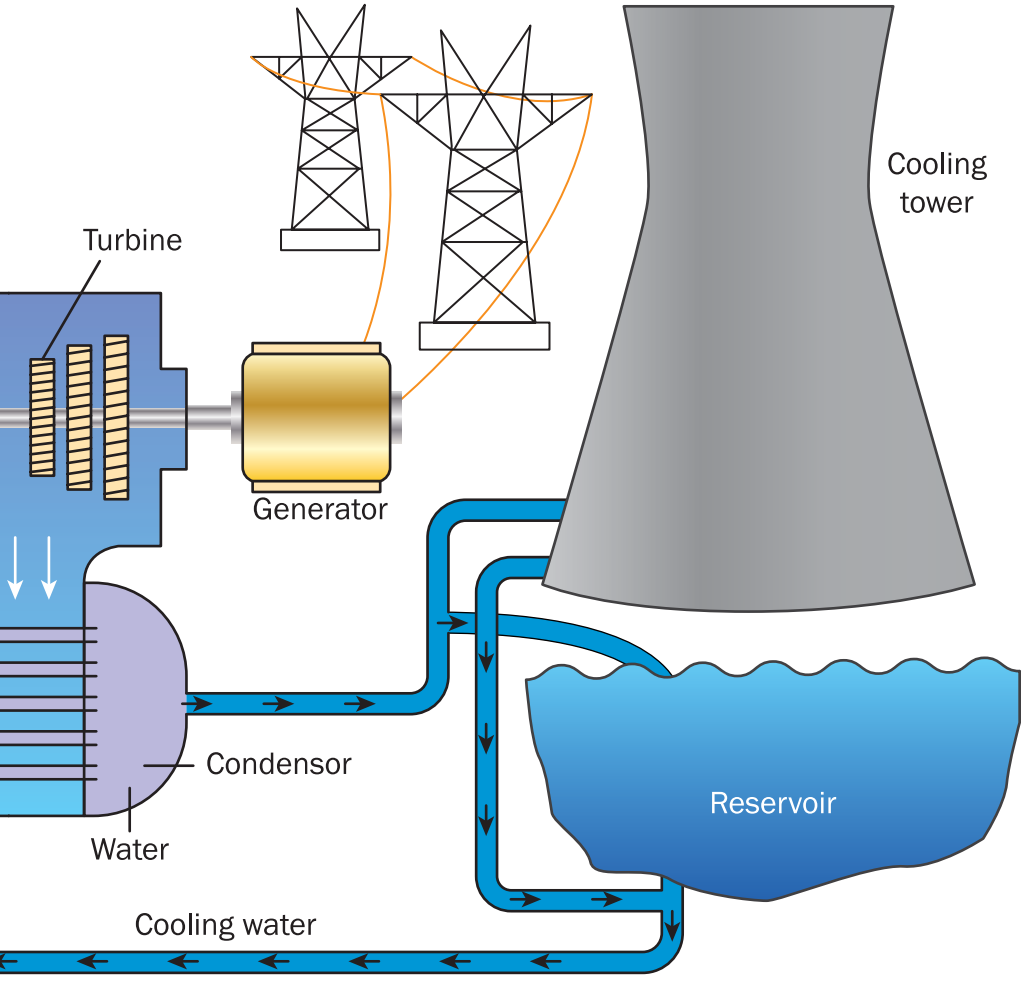
A specific type of uranium must be used to release nuclear energy. This material is known as uranium-235.

Useful nuclear energy is released from uranium-235 in a nuclear power plant. Look at the image closely to see how nuclear energy is transformed to electricity. The process called a chain reaction that releases energy from uranium-235 takes place in the reactor core. The fuel rods control the chain reaction. The containment shell helps prevent radioactivity from leaking into the environment outside. Radioactivity can kill or damage living organisms.



A nuclear power plant is designed to contain radioactivity. It generates electricity and slowly releases heat into the environment in a way that isn't harmful.

When atoms are split in a chain reaction, heat is generated. This heat converts liquid water into steam. The steam passes through a turbine, which powers a generator. The generator produces electricity, which is either stored or sent to power homes, businesses, and other structures. The steam then condenses back into liquid water. The water is cooled by pipes carrying water from outside the plant, usually from a river, lake, or sea.



The Costs of Nuclear Power

There are many costs and benefits in using nuclear energy. Some of the benefits were discussed on previous pages. One problem is that nuclear energy is nonrenewable. The materials that are the fuel for a nuclear reactor are rare. Also, nuclear power generates a lot of dangerous waste. Some of the material quickly breaks down into safe substances, but some remains radioactive for tens of thousands of years. This means that some materials need to be placed in sealed storage facilities that must be guarded or kept secret for centuries.

Nuclear material has been used to make weapons in addition to generating electricity. Nuclear bombs have devastating power. Because one nuclear weapon can do so much damage, governments have struggled to control access to nuclear materials and technology. A nation that claims to want nuclear technology only to provide electricity could also want it to develop weapons.



Nuclear power offers a reliable way of generating electricity without producing a lot of air pollution. What you see coming out of the cooling towers pictured here is steam.